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Learning to Draw ASCII Improves Spatial Reasoning in Language Models

Shiyuan Huang[*], **Li Liu**[*], **Jincheng He**, **Leilani H. Gilpin**
University of California, Santa Cruz   *Equal contribution   Corresponding author: lgilpin@ucsc.edu

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Project page

+--[Motivation]--+

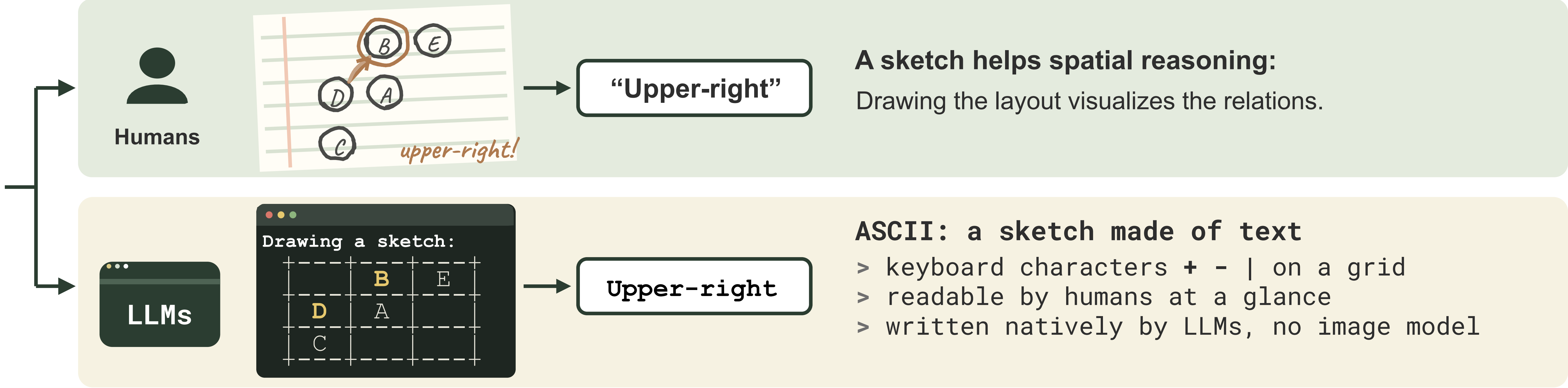
Humans sketch to reason about space. Can language models sketch too?

A spatial description

A is below B.
D is to the left of A.
C is at 6 o'clock from D.
E is to the northeast of A.
B is to the left of E.
A is at 1 o'clock from C.

Where is B relative to D?

An example of a spatial reasoning question.



```
Drawing a sketch:
+---+---+---+
|   | B | E |
+---+---+---+
| D | A |   |
+---+---+---+
| C |   |   |
+---+---+---+
```

A sketch helps spatial reasoning:
Drawing the layout visualizes the relations.

`ASCII: a sketch made of text`

```
> keyboard characters + - | on a grid
> readable by humans at a glance
> written natively by LLMs, no image model
```

+--[What we ask]---+

Four roles of an ASCII sketch

Representation
`[RQ1]` Can LLMs **construct** a layout from text, and **interpret** one?

Reasoning step
`[RQ2]` Does a sketch help them **answer** spatial questions?

Signal
`[RQ3]` Without ground truth, does **sketch–answer agreement** show which answers to trust?

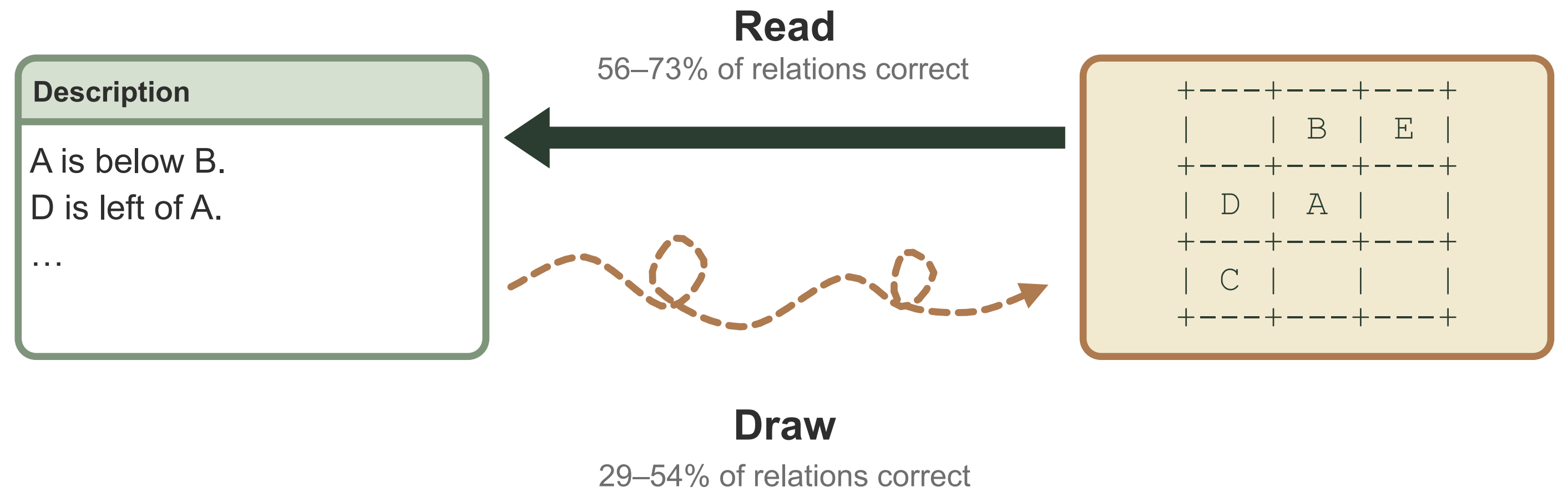
Training goal
`[RQ4]` Does **construction training** improve reasoning from text alone?

Our dataset, Text2Space: 20K spatial puzzles (2–8 objects, 3 ASCII styles), each paired with its ground-truth ASCII layout, so construction errors can be told apart from reasoning errors.

+--[Finding 1 · RQ1]-------------------+

Read-write asymmetry

LLMs read layouts well but struggle to construct them.



Model	Draw	Read
Qwen3-30B	29.2	55.8
Llama-3-70B	43.9	57.4
gpt-oss-20B	44.0	58.1
Qwen3-235B	46.0	59.7
GPT-4.1	53.6	72.5

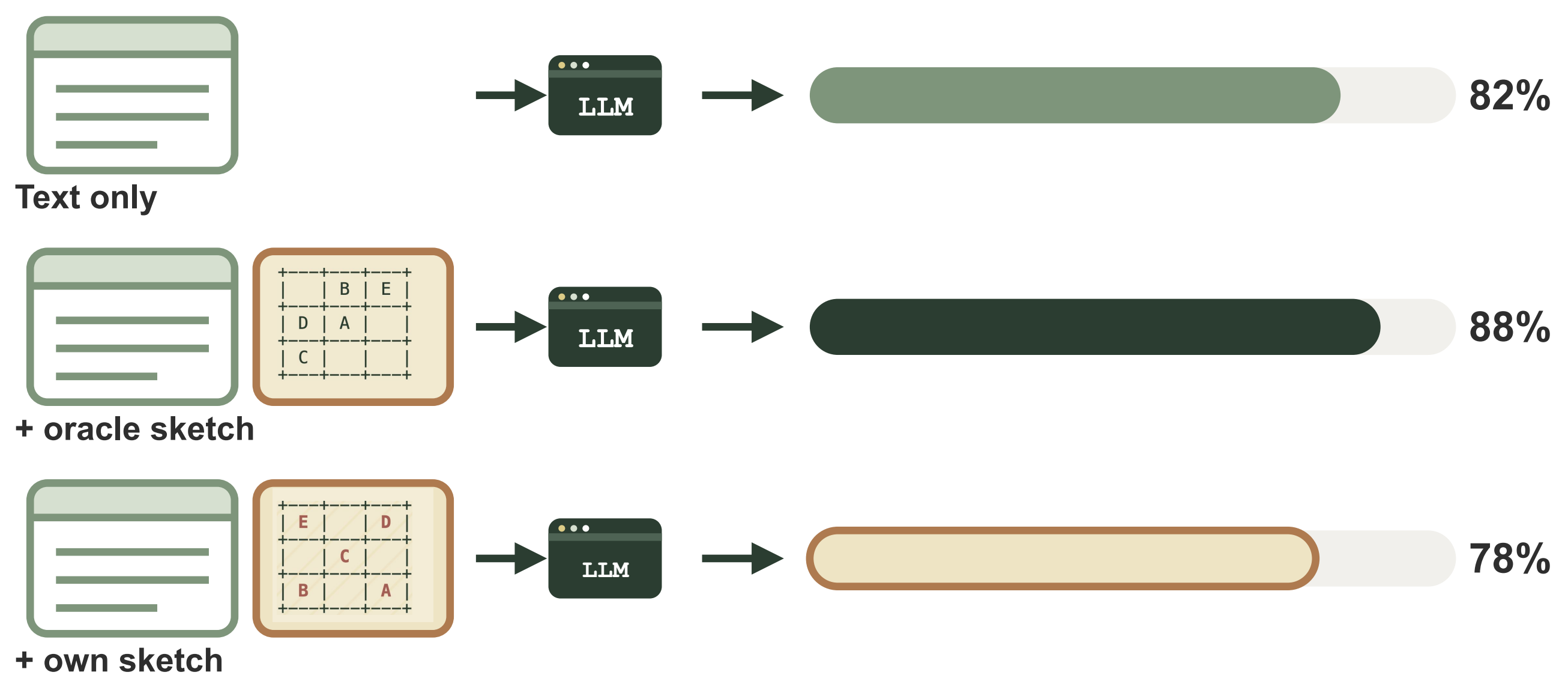
Relations correct (%). Draw: text → ASCII. Read: ASCII → text.

+--[Finding 2 · RQ2]--------------------+

Oracle sketch vs. own sketch

An oracle sketch helps, while the model's own sketch hurts: construction errors carry into the answer. Humans show the same gap (n = 11).

mean answer accuracy, 5 models



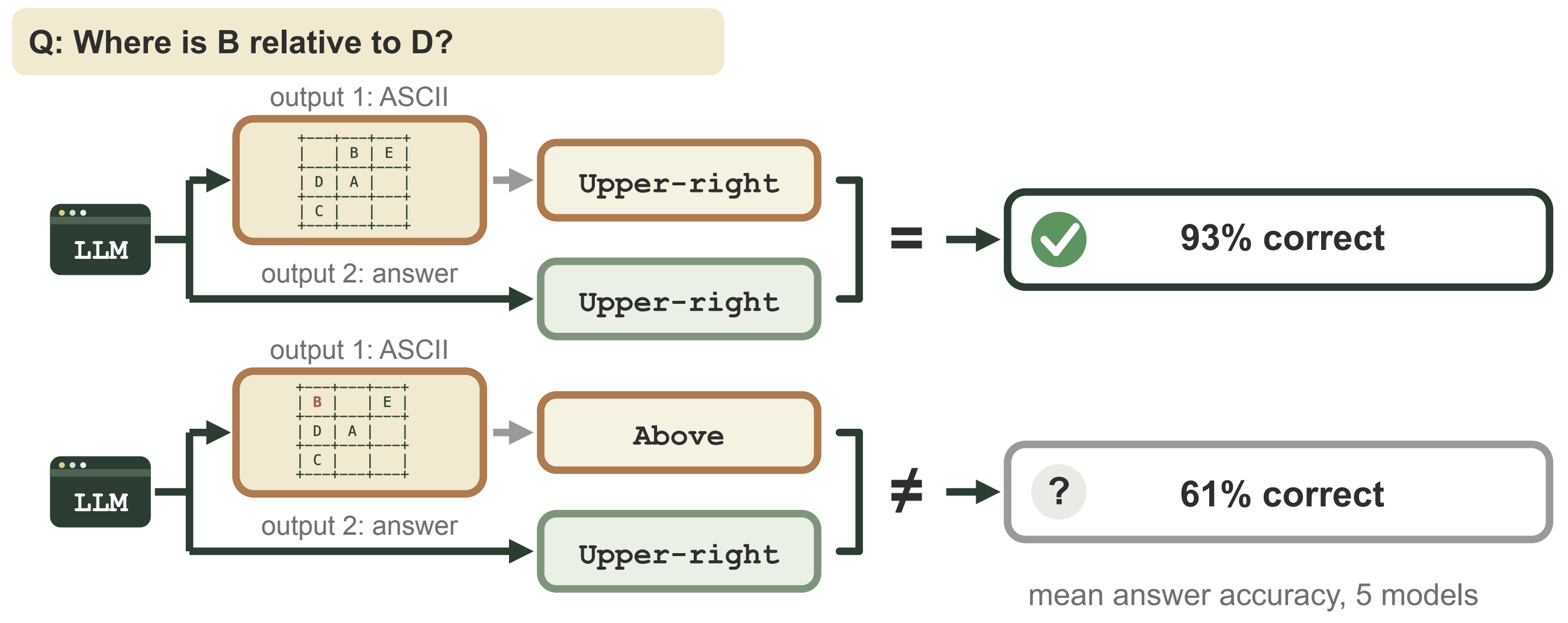
Model	Text only	+ oracle sketch	+ own sketch
Qwen3-30B	75.0	83.7	66.6
Llama-3-70B	71.7	78.4	66.9
gpt-oss-20B	93.3	97.0	91.9
Qwen3-235B	84.8	90.1	82.3
GPT-4.1	85.8	92.8	83.8

Answer accuracy (%). Oracle: the ground-truth ASCII is given. Own sketch: the model draws ASCII, then answers.

+--[Finding 3 · RQ3]------------------+

Agreement as a reliability signal

When the sketch and the answer agree, the answer is usually right. When they disagree, accuracy drops sharply.



mean answer accuracy, 5 models

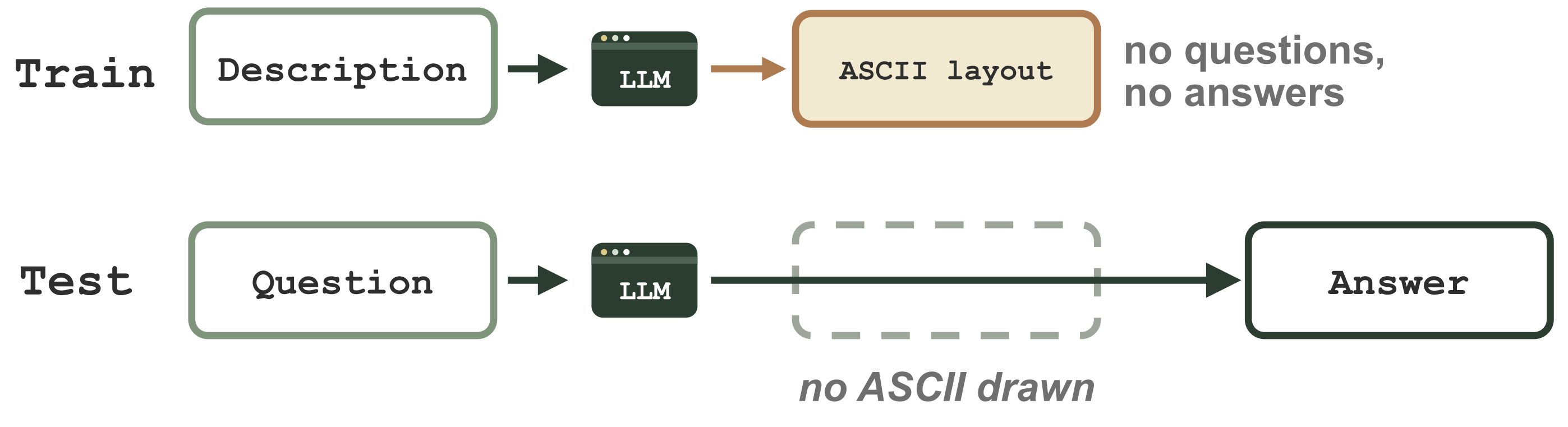
Model	Acc. if disagree	Acc. if agree	Agree rate
Qwen3-30B	56.2	88.5	46.1
Llama-3-70B	39.9	87.7	65.2
gpt-oss-20B	85.9	97.0	75.9
Qwen3-235B	61.5	95.0	67.8
GPT-4.1	62.4	94.9	74.2

Accuracy (%) when the sketch and the answer disagree or agree; agree rate = share of answers that agree (paper Tables 3, 10).

+--[Finding 4 · RQ4]-------------------+

ASCII training transfers to broader spatial reasoning tasks

Training only to draw improves reasoning from text alone, with no ASCII at test time, across models and external benchmarks.



Benchmark	Qwen3-30B	Llama-3-8B	Llama-3-70B
Text2Space	75.0 → **82.2**	42.4 → **61.5**	71.7 → **79.4**
StepGame[1, 2]	54.9 → **64.8**	44.8 → **46.3**	55.7 → **66.4**
bAbI-19[3]	17.8 → **34.8**	17.7 → **20.3**	56.3 → **60.7**
SpartQA[4]	26.5 → **69.8**	51.3 → **61.2**	59.2 → **72.7**

Answer accuracy (%), base → construction training (LoRA, 4,000 text-to-ASCII pairs); no ASCII at test time; external benchmarks are under zero-shot settings.

+--[Takeaway]--+

ASCII gives LLMs a sketchpad in text, and learning to draw on it can improve spatial reasoning.

`[1]` Auditable intermediate reasoning

ASCII is a text-native layout, so the intermediate spatial reasoning becomes visible and can be checked relation by relation.

`[2]` Accumulated error

A wrong scaffold misleads the answer: errors in a self-drawn layout propagate to the output, while an oracle layout improves it.

`[3]` Read-write asymmetry

Generation is harder than comprehension. Generation is the bottleneck, and training on it improves spatial reasoning beyond the training task.

`[4]` Agreement as a free check

The sketch and the answer are two outputs of one model. When they agree the answer is usually right; disagreement flags a probable error.

+--[References]-----------------------+

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